

BIHAR BOARD – 2019

SOCIAL SCIENCE

Time: 3hrs 15minutes

Max Marks: 100

SECTION –A

Objective Type Questions

Q 1. One card is drawn at random form a well shuffled deck of playing cards ,then the probability of getting an ace is –

- (A) $\frac{1}{4}$ (B) $\frac{1}{26}$ (C) $\frac{1}{13}$ (D) $\frac{4}{13}$

Ans. $\frac{1}{13}$

Q 2.If the ratio of areas of two similar triangles is 64:121, than the ratio of their corresponding sides is-

- (A) 8:11 (B) 8:12 (C) 12:14 (D) 11:8

Ans. 8:11

Q 3. If the means 6, 8, 9, x and 13 is 10 ,then the value of x is -

- (A) 12 (B) 13 (C) 14 (D) 15

Ans. 14

Q 4. $\cos 1^\circ \cos 3^\circ \dots \cos 90^\circ =$

- (A) 0 (B) 1 (C) -1 (D) $\sqrt{2}$

Ans. 0

Q 5. Pair of equations $x + 2y + 5 = 0$ and $-6y + 1 = 0$ have –

- (A) Unique solution (B) Infinite solution (C) No solution
(D) None of these

Ans. Unique solution

Q 6. Which of the following is not a quadratic equation?

- (A) $(x + 2)^2 = x(x^2 - 1)$ (B) $(x + 1)^2 = 2(x - 3)$
(C) $(x - 2) = (x + 2) = 5$ (D) $x^2 + \frac{1}{x^2} = 2$

Ans. $(x - 2) = (x + 2) = 5$

Q 7 . The length of a tangent from a point P at a distance 13cm from the centre of a circle is 12cm, then the radius of the circle is =

- (A) 6 cm (B) 12 cm (C) 9 cm (D) 5 cm

Ans. 5cm

Q 8. If roots of the quadratic equation $bx^2 + ax + c = 0$ are equal, then-

- (A) $b^2 - 4ac = 0$ (B) $a^2 - 4ac = 0$ (C) $c^2 - 4ab = 0$ (D) $a^2 - 4bc = 0$

Ans. $b^2 - 4ac = 0$

Q 9. If $\tan \theta = \frac{2}{4}$, then the value of $\sin \theta$ is=

- (A) $\frac{4}{5}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) $\frac{3}{5}$

Ans. $\frac{2}{3}$

Q 10 .PA and PB are two tangents drawn from an external point P to a circle .If PA =4 cm than the length of PB is –

- (A) 16 cm (B) 12 cm (C) 8 cm (D) 4 cm

Ans. 4cm

Q 11. Zeros of the quadratic polynomial x^2-2 are –

- (A) 2.2 (B) $-\sqrt{2}, \sqrt{2}$ (C) $-\sqrt{2}, -\sqrt{2}$ (D) $-2, -2$

Ans. $-\sqrt{2}, \sqrt{2}$

Q 12. Which of the following is not irrational?

- (A) $\sqrt{10}$ (B) $\sqrt{24}$ (C) $\sqrt{35}$ (D) $\sqrt{121}$

Ans. $\sqrt{121}$

Q13 . Value of $\tan^2 60^\circ$ is –

- (A) 1 (B) 3 (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{3}$

Ans. 3

Q 14. If the following quadratic polynomial whose sum of zeros is -3 and product is 2?

- (A) x^2+3x+2 (B) x^2+2x-3 (C) x^2-3x-2 (D) x^2-3x+2

Ans. $x^2 - 3x + 2$

Q15. In which quadrant does the point $(\sqrt{3}, -\sqrt{2})$ lie ?

- (A) 1st quadrant (B) 2nd quadrant (C) 3rd quadrant (D) 4th quadrant

Ans. 4th quadrant

Q 16. A line which intersects a circle at two distinct points is known as –

- (A) Chord (B) Tangent (C) Both (A) (B) (D) None of these

Ans. None of these

Q 17. If $5\cos\theta=3$, then the value of $3\tan\theta$ will be –

- (A) 3 (B) 5 (C) 4 (D) 7

Ans. 4

Q 18. If $p(x) = x^2 - 3x - 4$, then one of the zeros of $p(x)$ is –

- (A) 2 (B) 4 (C) 0 (D) 3

Ans. 4

Q 19 . If the length of the shadow of a vertical pole is $\sqrt{3}$ times the height of pole ,then the angle of elevation of sun is –

- (A) 45° (B) 30° (C) 75° (D) 60°

Ans. 30°

Q 20. If A, B, C are angles of a triangle, then $\sin\left(\frac{A+B}{2}\right)$ is equal to -

- (A) $\tan\frac{A}{2}$ (B) $\sin\frac{A}{2}$ (C) $\cos\frac{C}{2}$ (D) $\sec\frac{A}{2}$

Ans. $\cos\frac{C}{2}$

Q 21. If the equations $kx-5y=2$ and $6x+2y=7$ have no solutions, then -

- (A) $k=-10$ (B) $k=-5$ (C) $k=-6$ (D) $k=-15$

Ans. $k=-15$

Q 22. If p and q are two prime number, then their HCF is =

- (A) 2 (B) 0 (C) Either 1 or 2 (D) 1

Ans. 1

Q 23. The radius and height of a cone are r and h respectively, then its volume is =

- (A) $\frac{1}{2} \pi r^2 h$ (B) $\frac{4}{3} \pi r^2 h$ (C) $\frac{1}{3} \pi r^2 h$ (D) $\pi r^2 h$

Ans. $\frac{1}{3} \pi r^2 h$

Q 24. The 10th term of A.P: 54, 51, 48, 45.....is -

- (A) 27 (B) 30 (C) -27 (D) None of these

Ans. -27

Q 25. A man goes 24 m West, again he goes 10 m North. Now at what distance is the man from his initial point?

- (A) 34 cm (B) 17 cm (C) 26 cm (D) 28 m

Ans. 26cm

Q 26. $\cos(90^\circ - A) =$

- (A) $\cot A$ (B) $\sin A$ (C) $\tan A$ (D) $\sec A$

Ans. $\sin A$

Q27. If in two triangles ABC and PQR, $\angle A = \angle P$, $\angle B = \angle Q$, $\angle C = \angle R$, then –

- (A) $\triangle PQR \sim \triangle CAB$ (B) $\triangle PQR \sim \triangle BCA$ (C) $\triangle CBA \sim \triangle PQR$ (D) $\triangle ABC \sim \triangle PQR$

Ans. $\triangle ABC \sim \triangle PQR$

Q 28. The arithmetic mean of $x, x+3, x+6, x+9$ and $x+12$ is –

- (A) $x+6$ (B) $x+5$ (C) $x+7$ (D) $x+8$

Ans. $x+6$

Q 29 . The radius of a circle with area 38.5 cm^2 is

- (A) 7 cm (B) 3.5 cm (C) 10.5 cm (D) None of these

Ans. 3.5cm

Q 30. For an event E, which of the following is correct?

- (A) $P(E) > 1$ (B) $P(E) < 0$ (C) $P(E) = 1$ (D) $P(E) = -1$

Ans. $P(E) = 1$

Q 31. From a solid cube of edge 14 cm ,a spheres of largest volume is cut . Approximate volume of sphere is –

- (A) 359 cm³ (B) 1437 cm³ (C) 2874 cm³ (D) None of these

Ans. 1437 cm³

Q32. $\frac{\sin 18^\circ}{\cos 72^\circ} = ?$

- (A) 0 (B) 1 (C) -1 (D) ∞

Ans. 1

Q 33. LCM of two number a and 18 is 36 and HCF is 2,then the value of 'a' is

- (A) 2 (B) 3 (C) 4 (D) 1

Ans. 4

Q 34. If the of two linear equations in two variables is intersecting lines, than the number of solutions is –

- (A) Only one (B) No solutions (B) Infinite solutions (D) None of these

Ans. Infinite solutions

Q 35. Range of 24,15,22,13,9,10 and 30 will be -

- (A) 22 (B) 24 (C) 9 (D) 21

Ans. 21

Q36. Which of the following is equal to $\cot \theta$?

- (A) $\frac{\sin \theta}{\cos \theta}$ (B) $\frac{\cos \theta}{\sin \theta}$ (C) $\frac{1}{\sec \theta}$ (D) $\frac{1}{\sin \theta}$

Ans. $\frac{\cos \theta}{\sin \theta}$

Q 37. Distance between the points $(-5,7)$ and $(-1,3)$ is –

- (A) $2\sqrt{2}$ (B) $3\sqrt{2}$ (C) $4\sqrt{2}$ (D) $5\sqrt{2}$

Ans. $4\sqrt{2}$

Q38. Distance of the point $P(-6, 8)$ from the origin is –

- (A) 8 units (B) $2\sqrt{7}$ units (C) 6 units (D) 10 units

Ans. 10 units

Q 39. $\sin^2\theta + \cos^2\theta =$

- (A) 1 (B) 2 (C) 0 (D) -1

Ans. 1

Q 40. One of the side of an equilateral triangle is $2a$, then its height is –

- (A) $3a$ (B) $\sqrt{3}a$ (C) $\sqrt{3a^2}$ (D) $\frac{\sqrt{3}}{2}a$

Ans. $\sqrt{3}a$

Q 41. Which of the following cannot be the probability at an event ?

- (A) 3.5 (B) 0.2 (C) 0.3 (D) 60%

Ans. 3.5

Q 42. If $\sin\theta = \cos\theta$, then θ is equal to –

- (A) 45° (B) 30° (C) 90° (D) 60°

Ans. 45°

Q 43. How many cubes can be formed of side 2 cm from a cube of side 6 cm ?

- (A) 56 (B) 54 (C) 28 (D) 27

Ans. 27

Q 44. The ratio between the LCM and HCF of 5, 15 and 20 is –

- (A) 9:1 (B) 4:3 (C) 11:1 (D) 12:1

Ans. 12:1

Q 45. Radii of two cylinders are in the ratio 2:3 and their heights are in the ratio 5:3, then their volumes are in the ratio =

- (A) 27:20 (B) 20:27 (C) 4:9 (D) 9:20

Ans. 20:27

Q 46. The number of odd number between 0 and 50 is =

- (A) 26 (B) 25 (C) 27 (D) 24

Ans. 24

Q 47. The product of the smallest prime number and the smallest composite number is =

- (A) 10 (B) 6 (C) 8 (D) 4

Ans. 8

Q 48. The coordinates of the mid-point of line segment joining the points

P(-4,2) and Q(8,6) are =

- (A) (3,1) (B) (1,3) (C) (2,4) (D) (4,2)

Ans. (2,4)

Q 49. The coordinates of the mid –point of the line segment joining the points A(2,-4) and B(4,-2) is =

- (A) (6,-6) (B) (-1,-2) (C) (3,-3) (D) (-3,3)

Ans. (3, -3)

Q 50. Which term of the AP:2,6,10,14.....is 82 ?

- (A) 15° (B) 20° (C) 21° (D) 22°

Ans. 21

Section - B

Q 1. Convert $2.\overline{47}$ in the form of $\frac{p}{q}$.

Ans. Given decimal number is $2.\overline{47}$

Let, $x = 2.\overline{47}$ (i)

Now,

Multiplying the number by 100, we get

$$100x = 100 \times 2.\overline{47}$$

$$100x = 247.\overline{47} \text{(ii)}$$

Subtracting (i) from (ii), we get

$$100x - x = 247.\overline{47} - 2.\overline{47}$$

$$\Rightarrow 99x = 245$$

$$\therefore x = \frac{245}{99}$$

$$\text{Hence, } 2.\overline{47} = \frac{245}{99}$$

Q2 Solve the system of equations $2x+y=5$ and $3x+2y=8$ by cross –multiplication method.

Ans. Given, $2x + y = 5$
 $2x + y - 5 = 0$ (i)

$$3x + 2y = 8$$

$$3x + 2y - 8 = 0$$
(ii)

Now,

$$\frac{x}{1 \cdot -5} = \frac{y}{2 \cdot -8} = \frac{1}{2 \cdot -5 - 3 \cdot -8}$$

$$\Rightarrow \frac{x}{-5} = \frac{y}{-16} = \frac{1}{4}$$

$$\Rightarrow \frac{x}{-5} = \frac{y}{-16} = \frac{1}{4}$$

Taking,

$$\frac{x}{-5} = \frac{1}{4}$$

$$\Rightarrow x = 2 \times 1$$

$$\therefore x = 2$$

And, taking

$$\frac{y}{-16} = \frac{1}{4}$$

$$\Rightarrow y = 1 \times 1$$

$$\therefore y = 1$$

Q 3. Determine the AP whose 5th term is 15 and the sum of its 3rd and 8th terms is 34.

Ans. Let the first term of the A.P be a and the common difference be d .

We know, formula to find n th term is $a_n = a + (n - 1)d$

Now, according to question,

$$a_5 = 15$$

$$\Rightarrow a + (5 - 1)d = 15 \quad [\text{Given, fifth term is 15}]$$

$$\Rightarrow a + 4d = 15 \dots\dots\dots(i)$$

And,

Given, sum of 3rd and 8th terms is 34

$$a_3 + a_8 = 34$$

$$\Rightarrow a + (3 - 1)d + a + (8 - 1)d = 34$$

$$\Rightarrow a + 2d + a + 7d = 34$$

$$\Rightarrow a + 5d + a + 4d = 34$$

$$\Rightarrow a + 5d + 15 = 34 \dots\dots\dots[\text{Taking value from (i)}]$$

$$\Rightarrow a + 5d = 19$$

$$\therefore a_6 = 19 \dots\dots\dots[\because a_6 = a + (6 - 1)d]$$

Here, $a_5 = 15, a_6 = 19$.

$$\therefore d = a_6 - a_5 = 19 - 15 = 4$$

Now, putting value of d in (i), we get

$$a + 4 \times 4 = 15$$

$$\Rightarrow a = 15 - 16$$

$$\therefore a = -1$$

Hence, the required numbers of A.P are

$$a = -1$$

$$a_2 = -1 + 4 = 3$$

$$a_3 = -1 + (3-1)4 = 7$$

Q4. Using quadratic formula find the solutions of the equation $x^2 - 3x - 10 = 0$.

Ans. Given equation is $x^2 - 3x - 10 = 0$

Comparing it with quadratic equation $ax^2 + bx + c = 0$, we get

$a=1$, $b = -3$ and $c = -10$

Now, substituting the values in the formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \times 1 \times -10}}{2 \times 1}$$

$$\Rightarrow x = \frac{3 \pm \sqrt{9 + 40}}{2}$$

$$\Rightarrow x = \frac{3 \pm \sqrt{49}}{2}$$

$$\Rightarrow x = \frac{3 \pm 7}{2}$$

Either,

$$x = \frac{3+7}{2} = \frac{10}{2} = 5$$

or

$$x = \frac{3-7}{2} = \frac{-4}{2} = -2$$

$\therefore x = 5$ or -2

Q5. Prove that $5 - \sqrt{3}$ is an irrational number.

Ans. Given number is $5 - \sqrt{3}$

Let us assume that $(5 - \sqrt{3})$ is a rational number

Then, $(5 - \sqrt{3})$ can be written in the form $\frac{p}{q}$ where p and q are co- prime.

Therefore,

$$(5 - \sqrt{3}) = \frac{p}{q}$$

$$\Rightarrow -\sqrt{3} = \frac{p}{q} - 5$$

$$\Rightarrow -\sqrt{3} = \frac{p-5q}{q}$$

$$\Rightarrow \sqrt{3} = -\left(\frac{p-5q}{q}\right)$$

$$\Rightarrow \sqrt{3} = \frac{-p+5q}{q}$$

Here, $\frac{-p+5q}{q}$ is rational number, but $\sqrt{3}$ is an irrational number

Since, Rational number $\neq$ Irrational number

$\therefore$ Our assumption is incorrect

Hence, $5 - \sqrt{3}$ is not a rational number.

Q 6. Find a point on x axis which is equidistant from the points (7,6) and (-3,4).

Ans. Let the given points be P(7,6) and Q(-3,4) and C (x,0) be the point on x-axis.

According to question, C is equidistant from A and B.

$$\therefore AC=BC$$

We know that, the distance between two points (x_1, y_1) and (x_2, y_2) is

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

So,

Distance between A and C is

$$\begin{aligned} AC &= \sqrt{(x-7)^2 + (0-6)^2} \\ &= \sqrt{(x-7)^2 + 36} \end{aligned}$$

And, Distance between B and C is

$$\begin{aligned} BC &= \sqrt{(x-(-3))^2 + (0-4)^2} \\ &= \sqrt{(x+3)^2 + 16} \end{aligned}$$

Since, $AC = BC$

$$\therefore \sqrt{(x-7)^2 + 36} = \sqrt{(x+3)^2 + 16}$$

Squaring both the sides, we get

$$\begin{aligned} \left(\sqrt{(x-7)^2 + 36} \right)^2 &= \left(\sqrt{(x+3)^2 + 16} \right)^2 \\ \Rightarrow (x-7)^2 + 36 &= (x+3)^2 + 16 \\ \Rightarrow x^2 - 14x + 49 + 36 &= x^2 + 6x + 9 + 16 \\ \Rightarrow x^2 - 14x - x^2 - 6x &= 25 - 85 \\ \Rightarrow -20x &= -60 \\ \Rightarrow x &= \frac{60}{20} \\ \therefore x &= 3 \end{aligned}$$

Hence, the required point on x – axis is (3,0).

Q 7. In a $\triangle ABC$ whose is right angled, two points D and E are on sides CA and CB respectively prove that -

$$AE^2 + BD^2 = AB^2 + DE^2$$

Ans. Construction: Join AE, BC and DE

Proof: In triangle ACE

$$AE^2 = AC^2 + CE^2 \dots\dots\dots(i)$$

[By Pythagorean's theorem]

And, In triangle BCD

$$BD^2 = CD^2 + BC^2 \dots\dots\dots(ii) \quad [\text{By Pythagorean's theorem}]$$

Now, adding (i) and (ii), we get

$$\begin{aligned} AE^2 + BD^2 &= AC^2 + CE^2 + CD^2 + BC^2 \\ \Rightarrow AE^2 + BD^2 &= (AC^2 + BC^2) + (CE^2 + CD^2) \\ \therefore AE^2 + BD^2 &= AB^2 + DE^2 \end{aligned}$$

Hence proved.

Q 8. If $\tan\theta = \frac{4}{3}$, then find the value of $\frac{3\sin\theta + 2\cos\theta}{3\sin\theta - 2\cos\theta}$

Ans. Given, $\tan\theta = \frac{4}{3}$

$$\frac{3\sin \theta + 2\cos \theta}{3\sin \theta - 2\cos \theta}$$

$$\frac{3\sin \theta - 2\cos \theta}{3\sin \theta - 2\cos \theta}$$

Dividing both numerator and denominator by $\cos \theta$, we get

$$\frac{3\sin \theta + 2\cos \theta}{\cos \theta}$$

$$\frac{3\sin \theta - 2\cos \theta}{\cos \theta}$$

$$= \frac{\frac{3\sin \theta}{\cos \theta} + \frac{2\cos \theta}{\cos \theta}}{\frac{3\sin \theta}{\cos \theta} - \frac{2\cos \theta}{\cos \theta}}$$

$$= \frac{3\tan \theta + 2}{3\tan \theta - 2}$$

$$= \frac{3 \times \frac{4}{3} + 2}{3 \times \frac{4}{3} - 2}$$

$$= \frac{4 + 2}{4 - 2}$$

$$= \frac{6}{2} = 3$$

$$\therefore \frac{3\sin \theta + 2\cos \theta}{3\sin \theta - 2\cos \theta} = 3$$

Q 9. Calculate the volume of a right circular cone whose height is 28cm and radius of base is 21cm .

Ans. Given , radius (r) = 21cm

Height (h) = 28cm

We know,

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

$$\begin{aligned} &= \frac{1}{3} \times \frac{22}{7} \times (21)^2 \times 28 \\ &= 22 \times 21 \times 28 \\ &= 12936 \text{ cm}^3 \end{aligned}$$

Q 10. Establish a relation between x and y if the points (x,y),(1,2) and (7,0) are collinear.

Ans. Given that (x,y) , (1,2) and (7,0) are collinear points.

So, slope of any two points must be same.

$$\begin{aligned}\frac{y-2}{x-1} &= \frac{0-2}{7-1} \\ \Rightarrow \frac{y-2}{x-1} &= \frac{-2}{6} \\ \Rightarrow \frac{y-2}{x-1} &= \frac{-1}{3} \\ \Rightarrow 3(y-2) &= -1(x-1) \\ \Rightarrow 3y-6 &= -x+1 \\ \Rightarrow 3y+x &= 1+6 \\ \Rightarrow x+3y &= 7\end{aligned}$$

Hence , for given points to be collinear, x and y must be related as $x + 3y = 7$.

11.The HCF of two number is 145 and their LCM is 2175 .if one number is 725,find other number.

Ans. Given, HCF = 145

LCM = 2175

One number = 725

Let the other number be x.

We know,

Product of two numbers = LCM x HCF

$$\Rightarrow 725 \times x = 145 \times 2175$$

$$\Rightarrow x = \frac{145 \times 2175}{725}$$

$$\therefore x = 435$$

Therefore, the required number is 435.

Q12. If σ, β are the zeros of the quadratic polynomial $4x^2 - 4x + 1$, then find the value of $\frac{\sigma}{\beta} + \frac{\beta}{\sigma}$.

Ans. The given quadratic equation is

$$4x^2 - 4x + 1$$

Comparing to the general quadratic equation,

$$ax^2 + bx + c$$

We get

$$a = 4, b = -4, c = 1$$

Given that σ and β are zeroes of the polynomial.

According to the properties of the zeros,

$$\sigma + \beta = -\frac{b}{a} = -\frac{(-4)}{4} = \frac{4}{4} = 1$$

$$\sigma \cdot \beta = \frac{c}{a} = \frac{1}{4}$$

Then,

$$\begin{aligned}
 & \frac{\sigma}{\beta} + \frac{\beta}{\sigma} \\
 &= \frac{\sigma^2 + \beta^2}{\sigma\beta} \\
 &= \frac{(\sigma + \beta)^2 - 2\sigma\beta}{\sigma\beta} \\
 &= \frac{(4)^2 - 2 \times \frac{1}{4}}{\frac{1}{4}} \\
 &= \frac{16 - \frac{1}{2}}{\frac{1}{4}} \\
 &= \frac{31}{\frac{1}{4}} \\
 &= \frac{31}{1} \times \frac{4}{1} \\
 &= 62
 \end{aligned}$$

Q13. In $\triangle ABC$, D and E are points on the sides AB and AC respectively such that $DE \parallel BC$. If $AD=x$, $DB=x-2$, $AE=x+2$, $EC=x-1$, then find the value of x.

Ans. Given, $AD=x$, $DB=x-2$, $AE=x+2$, $EC=x-1$

Here, In $\triangle ABC$, $DE \parallel BC$,

$$\frac{AD}{DB} = \frac{AE}{EC} \dots\dots(i) \quad [\text{Basic Proportionality theorem}]$$

Now,

Substituting the values in (i), we get,

$$\frac{x}{x-2} = \frac{x+2}{x-1}$$

By cross multiplying, we get,

$$x(x-1) = (x+2)(x-2)$$

$$\Rightarrow x^2 - x = x^2 - 4$$

$$\Rightarrow x^2 - x - x^2 = -4$$

$$\therefore x = -4$$

Q14. Prove that –

$$\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta} = \left(\frac{1 + \sin \theta}{\cos \theta} \right)^2$$

Ans. L.H.S =

$$\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta}$$

$$= \frac{\frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}}{\frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta}}$$

$$= \frac{1 + \sin \theta}{1 - \sin \theta}$$

$$= \frac{\cos \theta}{1 - \sin \theta}$$

$$= \frac{1 + \sin \theta}{1 - \sin \theta} \times \frac{1 + \sin \theta}{1 + \sin \theta}$$

$$= \frac{(1 + \sin \theta)^2}{(1)^2 - (\sin \theta)^2}$$

$$= \frac{(1 + \sin \theta)^2}{1 - \sin^2 \theta}$$

$$= \frac{(1 + \sin \theta)^2}{\cos^2 \theta}$$

$$[\because \sin^2 \theta + \cos^2 \theta = 1]$$

$$= \left(\frac{1 + \sin \theta}{\cos \theta} \right)^2$$

$$= R.H.S$$

Q15. Two vertices of a triangle are (3,5) and (-7,4). If its centroid is (2,1) then find its vertex .

Ans. Let A(3,5) and B(-7,4) be the two vertices of a triangle and C(x,y) be third vertex. Centroid of a triangle is G(2,1).

We know that,

The coordinates of the centroid of a triangle = $\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}$

$$\Rightarrow (2,1) = \frac{3 + (-7) + x}{3}, \frac{-5 + 4 + y}{3}$$

$$\Rightarrow (2,1) = \frac{-4 + x}{3}, \frac{-1 + y}{3}$$

Now,

$$\frac{-4 + x}{3} = 2$$

$$\Rightarrow -4 + x = 6$$

$$\Rightarrow x = 6 + 4$$

$$\therefore x = 10$$

And

$$\frac{-1 + y}{3} = 1$$

$$\Rightarrow -1 + y = 3$$

$$\Rightarrow y = 3 + 1$$

$$\therefore y = 4$$

Hence, the co-ordinate of the third vertex is (10,4).

Q16. Using Euclid's division algorithm, find the HCF of 570 and 1425.

Ans. Given numbers are 570 and 1425

Here, 1425 > 570

According to Euclid division algorithm

$$a = bq + r$$

Here, a = 1425 and b = 570

So,

$$1425 = 570 \times 2 + 285$$

Since remainder $\neq 0$

$$570 = 285 \times 2 + 0$$

remainder = 0

Therefore, the H.C.F of given numbers is 285.

Q17. Find the surface area of a sphere whose diameter is 14cm.

Ans. Given,

Diameter of sphere (d) = 14cm

$$\therefore \text{Radius}(r) = \frac{d}{2} = \frac{14}{2} = 7\text{cm}$$

We know,

Surface area of sphere =

$$\begin{aligned} &4\pi r^2 \\ &= 4 \times \frac{22}{7} \times (7)^2 \\ &= 4 \times \frac{22}{7} \times 49 \\ &= 616 \end{aligned}$$

Hence, the surface area of sphere is 616cm^2 .

Q18. Evaluate the following –

$$\frac{5\sin^2 30^\circ + \cos^2 45^\circ + 4\tan^2 60^\circ}{2\sin 30^\circ \cos 60^\circ + \tan 45^\circ}$$

Ans.

$$\frac{5\sin^2 30^\circ + \cos^2 45^\circ + 4\tan^2 60^\circ}{2\sin 30^\circ \cos 60^\circ + \tan 45^\circ}$$

$$= \frac{5\left(\frac{1}{2}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + 4(\sqrt{3})^2}{2 \times \frac{1}{2} \times \frac{1}{2} + 1}$$

$$= \frac{\frac{5}{4} + \frac{1}{2} + 12}{\frac{1}{2} + 1}$$

$$= \frac{\frac{5+2+48}{4}}{\frac{1+2}{2}}$$

$$= \frac{\frac{55}{4}}{\frac{3}{2}}$$

$$= \frac{55}{4} \times \frac{2}{3}$$

$$= \frac{55}{6}$$

Q 19. $\triangle ABC$ is an isosceles triangle in which $AB=AC$. A circle is drawn inside the triangle such that it touches AB at E , BC at D and AC at F . Show that D is the midpoint of BC .

Ans. Given $\triangle ABC$ is an isosceles triangle where $AB = AC$.

We know that any two tangents drawn from an external point to a circle are equal in length.

$$\therefore AF = AE$$

Also, we have

$$AB = AC$$

So, $AB - AE = AC - AF$

$$\therefore BE = CF$$

Now,

$BE = BD$, $CD = CF$ [$\because$ Any two tangents drawn from an external point to a circle are equal in length]

$$\therefore BD = CD$$

Thus, it is proved that D is the mid- point of BC.

Q20. Find the area of a quadrant of a circle if its circumference is 44 cm.

Ans. Given,

Circumference of circle = 44cm

$$\Rightarrow 2\pi r = 44 \quad [\because \text{Circumference of circle} = 2\pi r]$$

$$\Rightarrow 2 \times \frac{22}{7} \times r = 44$$

$$\Rightarrow r = \frac{44 \times 7}{22 \times 2}$$

$$\therefore r = 7$$

Therefore, radius of circle = $r = 7$ cm

Now, we know

Area of quadrant =

$$\begin{aligned} & \frac{1}{4} \pi r^2 \\ &= \frac{1}{4} \times \frac{22}{7} \times (7)^2 \\ &= \frac{1}{4} \times \frac{22}{7} \times 49 \\ &= 38.5 \end{aligned}$$

Hence, the area of the quadrant of a circle is 38.5cm².

Q21. Two dice are thrown together, then find the probability of getting of same digit on both dice.

Ans.

When two dice are thrown together,

Total no. Of outcomes $n(S) = 6 \times 6 = 36$

Let A be the event of getting same digit, then

$A = (1,1), (2,2), (3,3), (4,4), (5,5)$ and $(6,6)$.

$\therefore n(A) = 6$

Now,

Probability of getting same digit on both dice $= P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6}$

Q 22.

Class interval	0-10	10-20	20-30	30-40	40-50
Frequency	8	12	6	15	7

Find mode.

Ans. Since 15 is the highest frequency, so the mode class interval is 30-40.

Now,

$$l = 30$$

$$f_1 = 15$$

$$f_0 = 6$$

$$f_2 = 7$$

$$h = 10$$

Using formula for mode,

Mode =

$$\begin{aligned}
 & l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h \\
 &= 30 + \left(\frac{15 - 6}{2 \times 15 - 6 - 7} \right) \times 10 \\
 &= 30 + \left(\frac{9}{30 - 13} \right) \times 10 \\
 &= 30 + \frac{9}{17} \times 10 \\
 &= 30 + 5.30 \\
 &= 35.30(\text{approximately})
 \end{aligned}$$

Q23.

Class interval	0-8	8-16	16-24	24-32	32-40	40-48
Frequency	10	20	14	16	18	22

Find mean.

Ans.

Class	Mid-point(x)	Frequency(f)	fx
0-8	4	10	40
8-16	12	20	240
16-24	20	14	280
24-32	28	16	448
32-40	36	18	648
40-48	44	22	968
		$\sum f = 100$	$\sum fx = 2624$

$$\begin{aligned}
 \therefore \text{Mean} &= \frac{\sum fx}{\sum f} \\
 &= \frac{2624}{100} \\
 &= 26.24
 \end{aligned}$$

Q24.If the product of two consecutive natural number is 20, then find the number.

Ans. Let the first number be x , then another number will be $(x + 1)$.

According to question,

Product of the two consecutive natural numbers = 20

$$x(x+1) = 20$$

$$\Rightarrow x^2 + x = 20$$

$$\Rightarrow x^2 + x - 20 = 0$$

$$\Rightarrow x^2 + 5x - 4x - 20 = 0$$

$$\Rightarrow x(x+5) - 4(x+5) = 0$$

$$\Rightarrow (x+5)(x-4) = 0$$

Either,

$$(x+5) = 0$$

$$\therefore x = -5$$

Or,

$$(x-4) = 0$$

$$\therefore x = 4$$

Since it is given that x is a natural number and -5 is not a natural number as it is a negative number.

So, $x = 4$

Now, First number = $x = 4$

Second number = $(x + 1) = 4 + 1 = 5$

Hence, the required numbers are 4 and 5.

Q25.Two coins are tossed together. Find the probability of getting at least one head .

Ans. When two different coins are tossed together,

The sample space, $S = \{HH, HT, TH, TT\}$

$$\therefore n(S) = 4.$$

Let E be the event of getting head at least one head.

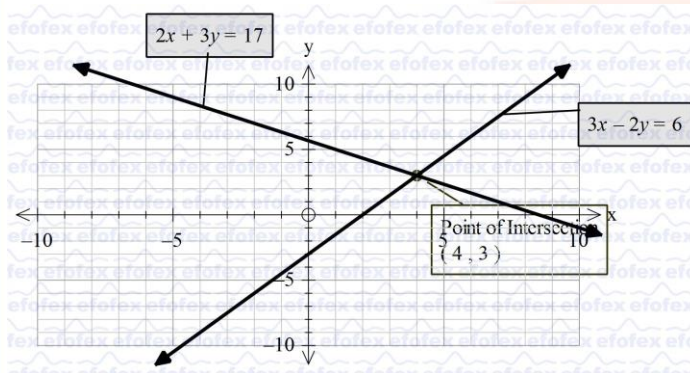
Then, $E = \{HH, HT, TH\}$

$$\therefore n(E) = 3$$

Therefore, probability of getting at least one head = $P(E) = \frac{n(E)}{n(S)} = \frac{3}{4}$

Q26. Draw the graph of pair of equations $2x+3y=17$ and $3x-2y=6$ and solve them .

Ans.



Q27. Two poles of equal heights are standing opposite to each other on either side of the road , which is 80m wide .From a point between them on the road ,the angles of elevation of the top of the poles are 60° and 30° respectively . Find the height of the poles and the distance of the point from the pole.

Ans. Let AB and CD be the two poles of equal height and their heights be h m.

BC be the 80 m wide road and let P be any point on the road.

Let,

CP be x m,

Then,

BP = $(80 - x)$

Also, $\angle APB = 60^\circ$ and $\angle DPC = 30^\circ$

Now,

In right angled triangle DCP,

$$\tan 30^\circ = \frac{CD}{CP}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x}$$

$$\Rightarrow h = \frac{x}{\sqrt{3}} \dots\dots\dots(i)$$

Again,

In right angled triangle ABP

$$\tan 60^\circ = \frac{CD}{CP}$$

$$\Rightarrow \sqrt{3} = \frac{h}{(80 - x)}$$

$$\Rightarrow h = \sqrt{3}(80 - x)$$

$$\Rightarrow \frac{x}{\sqrt{3}} = \sqrt{3}(80 - x) \quad [\text{From(i)}]$$

$$\Rightarrow x = 3(80 - x)$$

$$\Rightarrow x = 240 - 3x$$

$$\Rightarrow 4x = 240$$

$$\Rightarrow x = \frac{240}{4}$$

$$\therefore x = 60$$

Now, putting value of x in (i), we get

$$\text{Height of the pole (h)} = \frac{x}{\sqrt{3}} = \frac{60}{\sqrt{3}} = 20\sqrt{3}m$$

Hence, the distance of the point from the pole is 60 m and height of each pole is $20\sqrt{3}$ m.

Q28. Prove that the lengths of tangents drawn from an external point to a circle are equal.

Ans. Let two tangent PT and QT are drawn to circle of centre O as shown in figure.

Both the given tangents PT and QT touch to the circle at P and Q respectively.

To prove: length of PT = length of QT

Construction: - Draw a line segment, from centre O to external point T {touching point of two tangents} .

Now ΔPOT and ΔQOT

We know, tangent makes right angle with radius of circle.

Here, PO and QO are radii. So, $\angle OPT = \angle OQT = 90^\circ$

Now, it is clear that both the triangles ΔPOT and ΔQOT are right angled triangle and has a common hypotenuse OT as shown in figure.

Now,

ΔPOT and ΔQOT

$\angle OPT = \angle OQT = 90^\circ$

Common hypotenuse OT

And $OP = OQ$ [$\because$ OP and OQ are radii]

Therefore,

According to R - H - S rule of similarity

$\Delta POT \sim \Delta QOT$

So,

$$\frac{OP}{OQ} = \frac{TP}{QT} = \frac{OT}{OT}$$

$$\frac{PQ}{QT} = 1$$

$$\therefore PQ = QT$$

Hence proved

Q29. A train travels 360km at a uniform speed .If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey .Find the speed of the train.

Ans. Let the original speed of the train be x km/h.

Time taken to cover a distance of 360 km = $\frac{360}{x}$ hours.

When the speed increases by 5km/h,

New speed of the train = $(x+5)$ km/h.

Time taken to cover a distance of 360 km at new speed = $\frac{360}{x+5}$ hours.

Since, the train takes 1 hour less time,

So,

$$\frac{360}{x} - \frac{360}{x+5} = 1$$

$$\Rightarrow \frac{360(x+5) - 360x}{x(x+5)} = 1$$

$$\Rightarrow \frac{360(x+5-x)}{x^2+5x} = 1$$

$$\Rightarrow 360(5) = x^2 + 5x$$

$$\Rightarrow 1800 = x^2 + 5x$$

$$\Rightarrow x^2 + 5x - 1800 = 0$$

$$\Rightarrow x^2 + 45x - 40x - 1800 = 0$$

$$\Rightarrow x(x+45) - 40(x+45) = 0$$

$$\Rightarrow (x+45)(x-40)$$

$$\Rightarrow x = -45 \text{ or } 40$$

Since speed cannot be in negative form.

$\therefore x = 40$ km/hr.

Hence, the original speed of the train is 40 km/h.

Q 30. Prove that –

$$(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$$

Ans. L.H.S =

$$\begin{aligned} & (\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 \\ &= \sin^2 A + 2 \sin A \operatorname{cosec} A + \operatorname{cosec}^2 A + \cos^2 A + 2 \cos A \sec A + \sec^2 A \\ &= \sin^2 A + 2 \sin A \cdot \frac{1}{\sin A} + (1 + \cot^2 A) + \cos^2 A + 2 \cos A \cdot \frac{1}{\cos A} + (1 + \tan^2 A) \\ &= \sin^2 A + 2 + 1 + \cot^2 A + \cos^2 A + 2 + 1 + \tan^2 A \\ &= 6 + \sin^2 A + \cos^2 A + \tan^2 A + \cot^2 A \\ &= 6 + 1 + \tan^2 A + \cot^2 A \\ &= 7 + \tan^2 A + \cot^2 A \\ &= R.H.S \end{aligned}$$

Q31. PQ is a chord of length 8 cm of a circle of radius 5 cm, the tangents at P and Q intersect at a point T. Find the length of TP.

Ans. Given radius, $OP = OQ = 5$ cm

Length of chord, $PQ = 4$ cm

Since, $OT \perp PQ$,

$\therefore PM = MQ = 4$ cm [Perpendicular draw from the centre of the circle to a chord bisect the chord]

In right $\triangle OPM$,

$$\begin{aligned} OP^2 &= PM^2 + OM^2 \\ \Rightarrow 5^2 &= 4^2 + OM^2 \\ \Rightarrow OM^2 &= 25 - 16 \\ \Rightarrow OM^2 &= 9 \\ \therefore OM &= 3 \text{ cm} \end{aligned}$$

In right $\triangle PTM$,

$$PT^2 = TM^2 + PM^2 \dots\dots\dots(i)$$

$\angle OPT = 90^\circ$ [$\because$ Radius is perpendicular to tangent at point of contact]

In right $\triangle OPT$,

$$OT^2 = PT^2 + OP^2 \dots\dots\dots(ii)$$

From equations (i) and (ii), we get

$$\begin{aligned} OT^2 &= (TM^2 + PM^2) + OP^2 \\ \Rightarrow (TM + OM)^2 &= (TM^2 + PM^2) + OP^2 \\ \Rightarrow TM^2 + OM^2 + 2 \times TM \times OM &= TM^2 + PM^2 + OP^2 \\ \Rightarrow OM^2 + 2 \times TM \times OM &= PM^2 + OP^2 \\ \Rightarrow 32 + 2 \times TM \times 3 &= 42 + 52 \\ \Rightarrow 9 + 6TM &= 16 + 25 \\ \Rightarrow 6TM &= 32 \\ \Rightarrow TM &= 32 / 6 \\ \therefore TM &= 16 / 3 \end{aligned}$$

Now, substituting the value in Equation (i), we get

$$\begin{aligned} PT^2 &= TM^2 + PM^2 \\ \Rightarrow PT^2 &= \left(\frac{16}{3}\right)^2 + (4)^2 \\ \Rightarrow PT^2 &= \frac{256}{9} + 16 \\ \Rightarrow PT^2 &= \frac{256 + 144}{9} \\ \Rightarrow PT^2 &= \frac{400}{9} \\ \Rightarrow PT &= \sqrt{\frac{400}{9}} \\ \therefore PT &= \frac{20}{3} \end{aligned}$$

Therefore, the required length of tangent PT is $\frac{20}{3}$ cm.

Q32. Water is flowing with a speed of 10 km/hour in a canal 6 m wide and 1.5 m deep .How much area will it irrigate in 30 minutes, if 8 cm of standing water is needed for irrigation ?

Ans. Given, speed of water flow = 10 km/hour

Width of canal = 6m

Depth of canal = 1.5m

Now,

$$\text{Water flowing in canal in 30 minutes} = 10000 \times \frac{1}{2} = 5000m$$

[$\because$ Speed of water flow = 10km/hr]

Then,

$$\text{Volume of water flowing out in 30 minutes} = (5000 \times 6 \times 1.5) = 45000m^3$$

Let the area that can be irrigated in 30 minute be $A m^2$.

According to question,

$$\text{Volume of water required to irrigate the field} = \left(A \times \frac{8}{100} \right) m^3$$

Since, Volume of water flowing out of canal = Volume of water required to irrigate the field

So,

$$45000 = \left(A \times \frac{8}{100} \right)$$

$$\Rightarrow \frac{45000 \times 100}{8} = A$$

$$\therefore A = 562500m^2$$

Hence, $562500m^2$ of the field will be irrigated.

Q33. A well of diameter 3m is dug 14 m deep .the earth taken out of it has been spread evenly all around it in the shape of circular ring of width 4 m from an embankment. Find the height of the embankment.

Ans. Given,

Height of the well (h) = 14 m

Diameter of the well = 3 m

∴ Radius of the well (r) = 3/2 m

Now,

$$\begin{aligned}\text{Volume of the earth taken out of the well} &= \pi r^2 h \\ &= \frac{22}{7} \times \left(\frac{3}{2}\right)^2 \times 14 \\ &= 99m^3\end{aligned}$$

$$\text{Outer radius of the embankment} = R = \left(\frac{3}{2} + 4\right) = \frac{11}{2}m$$

Area of embankment = outer area - inner area

$$\begin{aligned}&= \pi R^2 - \pi r^2 \\ &= \frac{22}{7} \times \left(\frac{11}{2}\right)^2 - \frac{22}{7} \times \left(\frac{3}{2}\right)^2 \\ &= \frac{22}{7} \left(\frac{121}{4} - \frac{9}{4}\right) \\ &= \frac{22}{7} \times \frac{112}{4} \\ &= 88m^2\end{aligned}$$

∴ Height of the embankment

$$\begin{aligned}&\frac{\text{Volume}}{\text{Area}} \\ &= \frac{99}{88} \\ &= 1.125\end{aligned}$$

Hence, the height of the embankment is 1.125m.